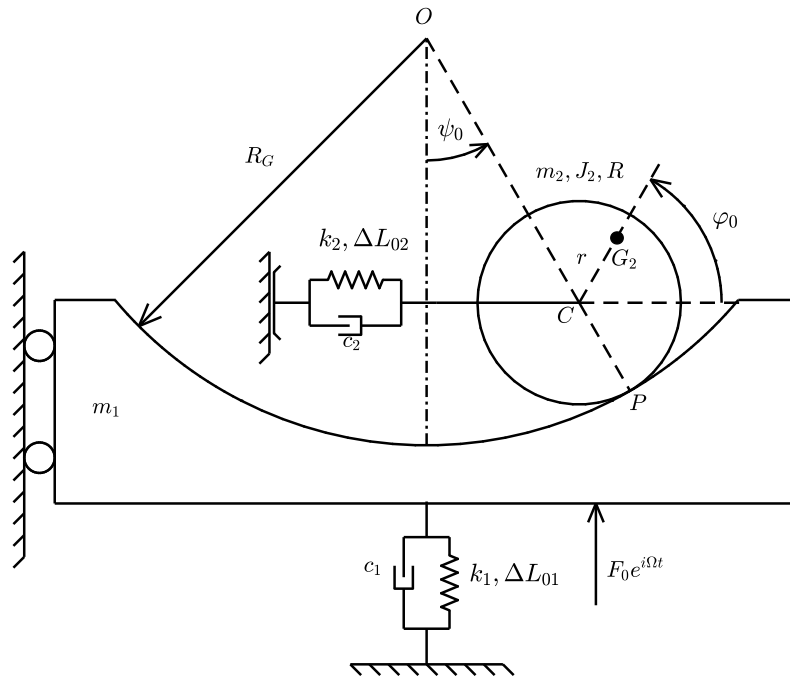


Problem 1:



$m_1 = 10 \text{ kg}$	$\Delta L_{01} = -0.0098 \text{ m}$
$m_2 = 5 \text{ kg}$	$\Delta L_{02} = -0.0014 \text{ m}$
$J_2 = 0.1 \text{ kgm}^2$	$c_1 = 10.0 \text{ Ns/m}$
$R_G = 1.0 \text{ m}$	$c_2 = 5.0 \text{ Ns/m}$
$R = 0.2 \text{ m}$	$\psi_0 = \pi/3$
$r = 0.15 \text{ m}$	$\varphi_0 = \pi/6$
$k_1 = 15000 \text{ N/m}$	
$k_2 = 15000 \text{ N/m}$	

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding on the upper circular side of the first rigid body; the distance $\overline{CG_2}$ between its center of mass G_2 and the disk center C is r .

The students are requested to:

0. Write on top of each page their surname, name, Person Code and SNxxxx code (see below)
1. Write the equations of motion of the system linearized around the given equilibrium configuration (provide this answer on paper: Jacobian matrices and second-order stiffness terms)
2. Compute the natural frequencies and modal shapes (provide this answer on paper, copying the values from MATLAB);
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic vertical force $F_0 e^{i\Omega t}$. Output: the vertical component of the displacement of point C and the horizontal component of the displacement of point G_2 . Save the graphical results as SNxxxx1.fig and SNxxxx2.fig, with SNxxxx your identifier, see below;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic vertical force $F_0 e^{i\Omega t}$. Output: the normal and tangential components of the contact forces in point P. Save the graphical result as SNxxxx3.fig and SNxxxx4.fig.
5. Compute the static preloads ΔL_{01} and ΔL_{02} when $\psi_0 = \pi/6$ and $\varphi_0 = \pi/3$ (provide the answer on paper).

The answer to points 1, 2 and 5 shall be provided on paper (for point 2 please copy on paper the values of the natural frequencies and eigenmodes produced by the MATLAB script); the answer to points 3 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions below.

You must upload one single archive file for both problem 1 and problem 2, which should contain your MATLAB scripts and all the .fig files you want to deliver as part of your solution.

- i.) Accepted formats for the archive files are: .7z .rar .zip;
- ii.) The file must be named SNxxxx with “S” the first letter in your surname, “N” the first letter in your name (in case of multiple surnames/names just consider the first one), “xxxx” the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the name of the file should be: BS5678.ext;
- iii.) The file must be uploaded as an assignment on WeBeeP, the name of the assignment is **Test_05_July_2022**;
- iv.) the deadline to upload the assignment will be announced during the written exam.

Your identifier: SNxxxx with S the first letter of your surname, N the first letter of your name and xxxx the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the identifier BS5678

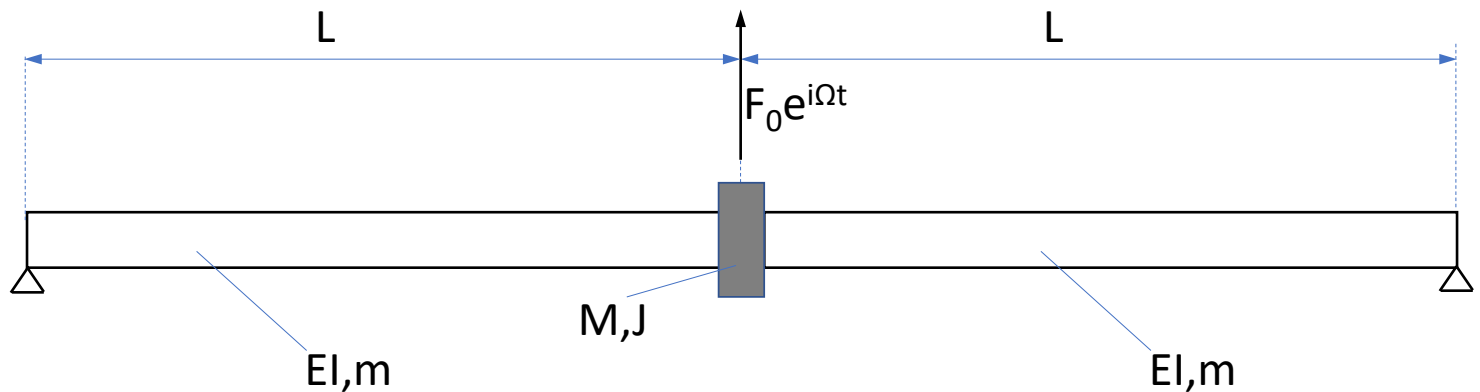
Please be aware that your answers to the test will not be considered in case:

- a. the information stated at point 0 above is missing;
- b. your archive file is uploaded in the wrong folder
- c. the name of the archive file is different from the identifier SNxxxx described above;

For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one.

Problem 2:

Consider the bending motion of a beam with uniform section and material properties having length $2L$. Both ends of the beam are simply supported. In the central section, a rigid body with mass M and moment of inertia J is rigidly attached to the beam.



DATA:

$$EI=4e5 \text{ Nm}^2$$

$$m=10 \text{ kg/m}$$

$$L=2 \text{ m}$$

$$M=20 \text{ kg}$$

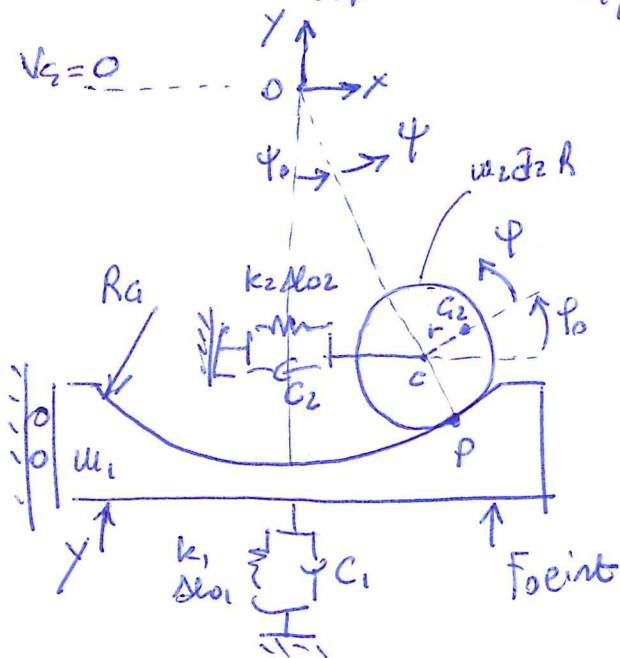
$$J=0.6 \text{ kgm}^2$$

- 1) Write the boundary conditions and the expression of the square matrix of coefficients H of the linear homogeneous system defining the natural frequencies of the system. Deliver this part of the solution on paper.
- 2) Compute the natural frequencies of the system in the frequency range 0-200 Hz and the related modal shapes. Deliver the values of the natural frequencies on paper and include in your archive file one .fig file for each modal shape. The .fig files shall be named SNxxxxP2i with SNxxxx your identifier (see below) and $i=1,2,3$ (use as many as needed);
- 3) Plot the Bode diagram (log y scale) of the magnitude of the FRF considering:
 - Non-dimensional damping factor: 0.04 for all modes
 - input: force having unit magnitude applied in the centre of beam as shown in the figure;
 - output: vertical vibration in the centre of beam;The Bode diagram shall be delivered as a .fig file named SNxxxxP2FRF included in your archive.

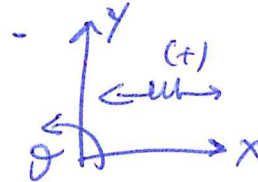
The answer to points 1 shall be provided on paper; the answer to points 2 – 3 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions reported for problem 1.

DYNAMICS OF MECHANICAL SYSTEMS

Written Test July 05th, 2022



- 2 rigid bodies (6 dof)
- 2 carts + rolling without sliding condition = 4 dof
- 2 dof

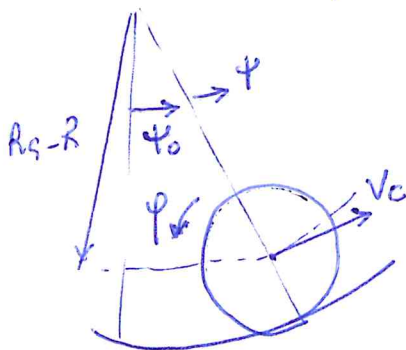


$$\underline{x} = \begin{Bmatrix} x \\ \psi \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} J_2 \dot{\psi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 c_i \Delta l_i^2; \quad Vel = \frac{1}{2} \sum_{i=1}^2 k_i \Delta l_i^2; \quad V_g = \sum_{i=1}^2 m_i g h_{ci}$$

$$\delta W = F_{0, \text{cent}} \delta y$$



$$v_c = (R_s - R) \dot{\psi} = -R \dot{\psi} \rightarrow \dot{\psi} = -\frac{R_s - R}{R} \dot{\psi}$$

$$\psi = -\frac{R_s - R}{R} \psi + \text{const.}$$

$$\begin{cases} \psi = 0 \\ \dot{\psi} = 0 \end{cases} \rightarrow 0 = -\frac{R_s - R}{R} \psi + \text{const} \rightarrow \text{const} = 0$$

$$\rightarrow \boxed{\psi = -\frac{R_s - R}{R} \psi = -\frac{\bar{R}}{R} \psi; \quad \bar{R} = R_s - R}$$

Non linear kinematic relationships:

$$x_c = \bar{R} \sin(\psi_0 + \psi)$$

$$y_c = y - \bar{R} \cos(\psi_0 + \psi)$$

$$x_{s2} = \bar{R} \sin(\psi_0 + \psi) + r \cos(\psi_0 + \psi) = \bar{R} \sin(\psi_0 + \psi) + r \cos(\psi_0 - \frac{\bar{R}}{R} \psi)$$

$$y_{s2} = y - \bar{R} \cos(\psi_0 + \psi) + r \sin(\psi_0 + \psi) = y - \bar{R} \cos(\psi_0 + \psi) + r \sin(\psi_0 - \frac{\bar{R}}{R} \psi)$$

$$\Delta L_2 = + \Delta x_c = x_c - x_{c0}$$

$$\frac{\partial x_{s2}}{\partial \psi} = \bar{R} \cos(\psi_0 + \psi) + \frac{r\bar{R}}{R} \sin(\psi_0 - \frac{\bar{R}}{R} \psi)$$

$$\frac{\partial y_{s2}}{\partial \psi} = \bar{R} \sin(\psi_0 + \psi) - \frac{r\bar{R}}{R} \cos(\psi_0 - \frac{\bar{R}}{R} \psi)$$

$$[u_{PH}] = \text{diag}(u_1, u_2, u_2, J_2) \quad ; \quad \underline{y}_n = \{y \quad x_{s2} \quad x_{s2} \quad \psi\}^T$$

$$[C_{PH}] = \text{diag}(C_1, C_2) \quad ; \quad [k_{PH}] = \text{diag}(k_1, k_2) \quad ; \quad \underline{y}_k = \underline{y}_c = \{\Delta u_1, \Delta u_2\}^T$$

$$\underline{F} = \{F_0 e^{i\omega t}\} \quad ; \quad \underline{y}_d = \{y\}$$

$$\underline{\Lambda}_{\text{gen}} = \begin{bmatrix} 1 & 0 \\ 0 & \bar{R} \cos \psi_0 + \frac{r\bar{R}}{R} \sin \psi_0 \\ 1 & \bar{R} \sin \psi_0 - \frac{r\bar{R}}{R} \cos \psi_0 \\ 0 & -\frac{\bar{R}}{R} \end{bmatrix} \quad \underline{\Lambda}_b = \begin{bmatrix} 1 & 0 \\ 0 & \bar{R} \cos \psi_0 \end{bmatrix}$$

$$\underline{\Lambda}_d = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

h_{s1} depends linearly from $y \rightarrow [k_{s1}] = \phi$

$$h_{s2} = y_{s2}$$

$$\frac{\partial^2 h_{s2}}{\partial \psi^2} = \bar{R} \cos(\psi_0 + \psi) - r \left(\frac{\bar{R}}{R} \right)^2 \sin(\psi_0 - \frac{\bar{R}}{R} \psi)$$

$$[k_{s2}] = u_2 g \begin{bmatrix} 0 & 0 \\ 0 & \bar{R} \cos \psi_0 - r \left(\frac{\bar{R}}{R} \right)^2 \sin \psi_0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta u_2}{\partial \psi^2} = -\bar{R} \sin(\psi_0 + \psi)$$

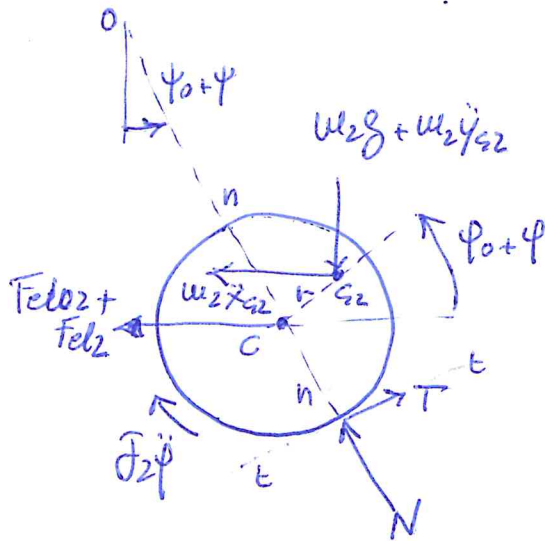
$$[k_{el, \pi_2}] = k_2 \Delta l_{02} \begin{bmatrix} 0 & 0 \\ 0 & -\bar{R} \sin \psi_0 \end{bmatrix}$$

• FRF: input $F_0 \sin t_0$

outputs: $x_{s2} \approx \gamma + (\bar{R} \sin t_0) \psi$

$$x_{s2} \approx (\bar{R} \cos \psi_0 + \frac{r\bar{R}}{R} \sin \psi_0) \psi$$

N_{dyu}, T_{dyu} :



$$\sum F_{n-n}^{Disk} = 0:$$

$$N + (F_{el02} + F_{el2}) \sin(\psi_0 + \psi) - (u_2 g + u_2 \ddot{y}_{s2}) \cos(\psi_0 + \psi) + u_2 \ddot{x}_{s2} \sin(\psi_0 + \psi) = 0$$

$$N + F_{el02} \sin t_0 + F_{el2} \psi \cos \psi_0 + F_{el2} \sin t_0 + F_{el2} \cos t_0 \psi - u_2 g \cos t_0 + u_2 g \psi \sin t_0 - u_2 \ddot{y}_{s2} \cos t_0 + u_2 \ddot{y}_{s2} \psi \sin t_0 + u_2 \ddot{x}_{s2} \sin t_0 + u_2 \ddot{x}_{s2} \psi \cos t_0 = 0$$

$$N_{sT} = -F_{el02} \sin t_0 + u_2 g \cos t_0$$

$$N_{dyu} = -F_{el2} \sin t_0 + u_2 \ddot{y}_{s2} \cos t_0 - u_2 \ddot{x}_{s2} \sin t_0 - \underbrace{(F_{el02} \cos t_0 + u_2 g \sin t_0) \psi}_{(*)}$$

$$\sum F_{t-t}^{Disk} = 0$$

$$T - (F_{el02} + F_{el2}) \cos(\psi_0 + \psi) - u_2 \ddot{x}_{s2} \cos(\psi_0 + \psi) - (u_2 g + u_2 \ddot{y}_{s2}) \sin(\psi_0 + \psi) = 0$$

$$T - F_{el02} \cos \psi_0 + F_{el2} \psi \sin \psi_0 - F_{el2} \cos \psi_0 + F_{el2} \psi \cos \psi_0 - u_2 \ddot{x}_{s2} \cos t_0 + u_2 \ddot{x}_{s2} \psi \sin \psi_0 - u_2 g \sin t_0 - u_2 g \psi \cos t_0 - u_2 \ddot{y}_{s2} \sin t_0 - u_2 \ddot{y}_{s2} \psi \cos t_0$$

$$T_{sT} = F_{el02} \cos \psi_0 + u_2 g \sin t_0$$

$$T_{dyu} = F_{el2} \cos t_0 + u_2 \ddot{x}_{s2} \cos t_0 + u_2 \ddot{y}_{s2} \sin t_0 + \underbrace{(u_2 g \cos t_0 - F_{el02} \sin t_0) \psi}_{(*)}$$

$$F_{el2} = (k_2 + i\Omega C_2) \Delta L_2, \Delta L_2 \approx (\bar{R} \cos \psi_0) \psi$$

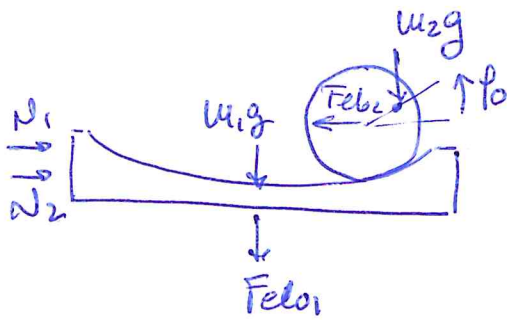
$$\ddot{x}_{s2} = -\Omega^2 x_{s2}, x_{s2} \approx (\bar{R} \cos \psi_0 + \frac{r\bar{R}}{R} \sin \psi_0) \psi$$

$$\ddot{y}_{s2} = -\Omega^2 y_{s2}, y_{s2} \approx \gamma + (\bar{R} \sin \psi_0 - \frac{r\bar{R}}{R} \cos \psi_0) \psi$$

- the 2nd order terms have been simplified

- (*) additional linear terms not mandatory

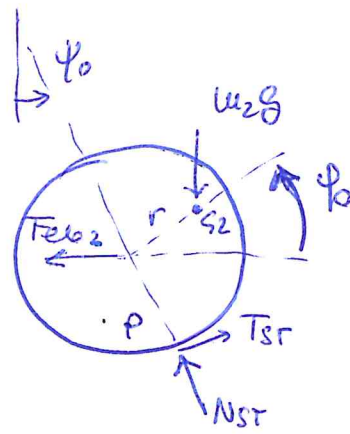
• static preloads:



$$\sum F_y = 0: m_1g + m_2g + F_{elo1} = 0$$

$$F_{elo1} = -m_1g - m_2g$$

$$\Delta l_{o1} = -\frac{(m_1 + m_2)g}{k_1}$$



$$\sum M_p^{\text{disk}} = 0:$$

$$F_{elo2} R \cos \phi_0 - m_2g (r \cos \phi_0 - R \sin \phi_0) = 0$$

$$F_{elo2} = m_2g \frac{r \cos \phi_0 - R \sin \phi_0}{k_2 R \cos \phi_0}$$

$$\Delta l_{o2} = \frac{F_{elo2}}{k_2}$$